

Python Optimization Tools Comparison

Tool	Problem Type	Algorithm	Typical Speed	Remarks
scipy.linprog	LP	HiGHS	< 0.1s @ 1000var	Fast, reliable
scipy.minimize (SLSQP)	QP/NLP	SQP	< 1s @ 500var	General purpose
scipy.minimize (L-BFGS-B)	Unconstrained	Quasi-Newton	< 0.5s @ 100dim	Large-scale
cvxpy + ECOS	Convex	Interior Point	< 1s @ 1000var	Modeling friendly
cvxpy + MOSEK	SDP/SOCP	Interior Point	< 1s @ 500var	Commercial solver